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Sensorimotor dysfunction following anterior cruciate ligament injury (Part 2): How can clinicians rehabilitate it?

Authors

Thilina N. Vitharana, PT, BSc, MSc [1, 2], Enda King, PT, PhD [3, 4], Neil Welch, PhD [1], Brian Devitt, MD, PhD [1,2], Kieran Moran, PhD [2, 5]

[1] Sports Medicine Department, UPMC Sports Surgery Clinic, Dublin, Ireland

[2] School of Health and Human Performance, Dublin City University, Dublin, Ireland

[3] Aspetar, Orthopaedic and Sports Medicine Hospital, FIFA Medical Centre of Excellence, Doha, Qatar

[4] Department of Life Sciences, Roehampton University, London, UK

[5] Insight Centre for Data Analytics, Dublin City University, Dublin, Ireland

Corresponding author: Thilina N Vitharana PT, UPMC Sports Surgery Clinic, Santry, Dublin 9, Ireland (email: vitharanat@upmc.ie) @Thilina_Physio

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Abstract

Background. Sensorimotor dysfunction following anterior cruciate ligament (ACL) injury and subsequent ACL reconstruction (ACLR) may lead to errors in motor planning and execution. Dysfunction occurs even after athletes have completed rehabilitation. Therefore, clinicians need to implement strategies to improve how sensorimotor dysfunction is addressed during rehabilitation.

Clinical question. How can clinicians improve their rehabilitation of sensorimotor dysfunction following ACLR? The aim of this clinical commentary is to review methods that may improve rehabilitation by targeting sensorimotor dysfunction following ACL injury.

Key results. Rehabilitation should focus on two priorities: (1) improving peripheral and central efferent function, and (2) improving somatosensory function and reducing reliance on the visual-motor system.

Clinical application. Somatosensory function can improve with proprioceptive training but should be implemented within the first six weeks of injury/surgery to minimise the chance of increased reliance on the visual-motor system. Methods to increase the complexity of proprioceptive training includes varying the: (1) type of task (e.g. jumping, decelerating, etc), (2) the visual information used during the task, (3) the cognitive loading during the task, and (4) perturbations. Visual distraction training may be useful for challenging an athlete during sports-specific training and reduce the reliance on the visual-motor system. Improving peripheral and central efferent function involves using regular cryotherapy and transcutaneous electrical nerve stimulation (TENS) in the early stages to minimise pain and improve muscle recruitment. Throughout rehabilitation, strength training, neuromuscular electrical stimulation (NMES) and surface EMG biofeedback methods improve central and peripheral efferent function when prescribed at a high intensity.

Key words

Anterior cruciate ligament, sensorimotor dysfunction, treatment, rehabilitation, afferent, efferent

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Introduction

Sensorimotor dysfunction can persist after rehabilitation, leading to errors in motor planning and execution,[25,56] as described in Part 1.[61] Rehabilitation following ACLR focuses on: (1) achieving a 'calm' knee joint (full ROM and no effusion) early post-surgery, (2) regaining strength and power, (3) introducing linear and multi-directional running, and (4) reintroducing sport.[60] With lingering sensorimotor dysfunction, it is possible that clinicians may not be using effective strategies to target these specific deficits. The aim of this clinical commentary is to review methods that may improve rehabilitation by targeting sensorimotor dysfunction following ACL injury and ACLR. We include evidence-based suggestions that are driven by our clinical practice in a sports medicine facility and mechanistic research. Key definitions are listed in Table 1.

Table 1. Key terms related to sensorimotor dysfunction

Term	**Definition**
Sensorimotor	A system integral to motor control within the body that encompasses afferent, efferent and central processing pathways
Efferent	Includes pathways that takes neural activity to the muscles to create movement
Central efferent function	Focuses on efferent activity within the brain
Peripheral efferent function	Focuses on efferent activity in the descending motor column and spinal reflexive activity
Afferent	Includes pathways that takes neural activity from the sensory systems to the brain
Somatosensory	Sensory system focused on perception of position, movement, touch pressure and pain
Proprioception	A function of the somatosensory system that relates to the body's ability to sense position of its body segments in space

Clinical question

How can clinicians improve their rehabilitation of sensorimotor dysfunction following ACL injury? Dysfunction within the afferent and efferent pathways following ACL injury and ACLR are summarised in Figure 1.[61] The two events that are pertinent to clinicians are: (1) reduced efferent function, and (2) increased reliance on the visual-motor system (figure 1). The reduced efferent function results from reduced central and peripheral efferent factors. Increased reliance on the

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visual-motor system occurs in response to altered somatosensory function, namely altered proprioceptive information following the damage to proprioceptors within the ACL. Therefore, clinicians should focus their rehabilitation on the following areas to effectively manage sensorimotor dysfunction (figure 1):

1. Improving peripheral and central efferent function
2. Improving somatosensory function and reducing reliance on the visual-motor system

The following sections review the methods of rehabilitation targeting the above priorities and providing guidelines for how clinicians may implement these in practice.

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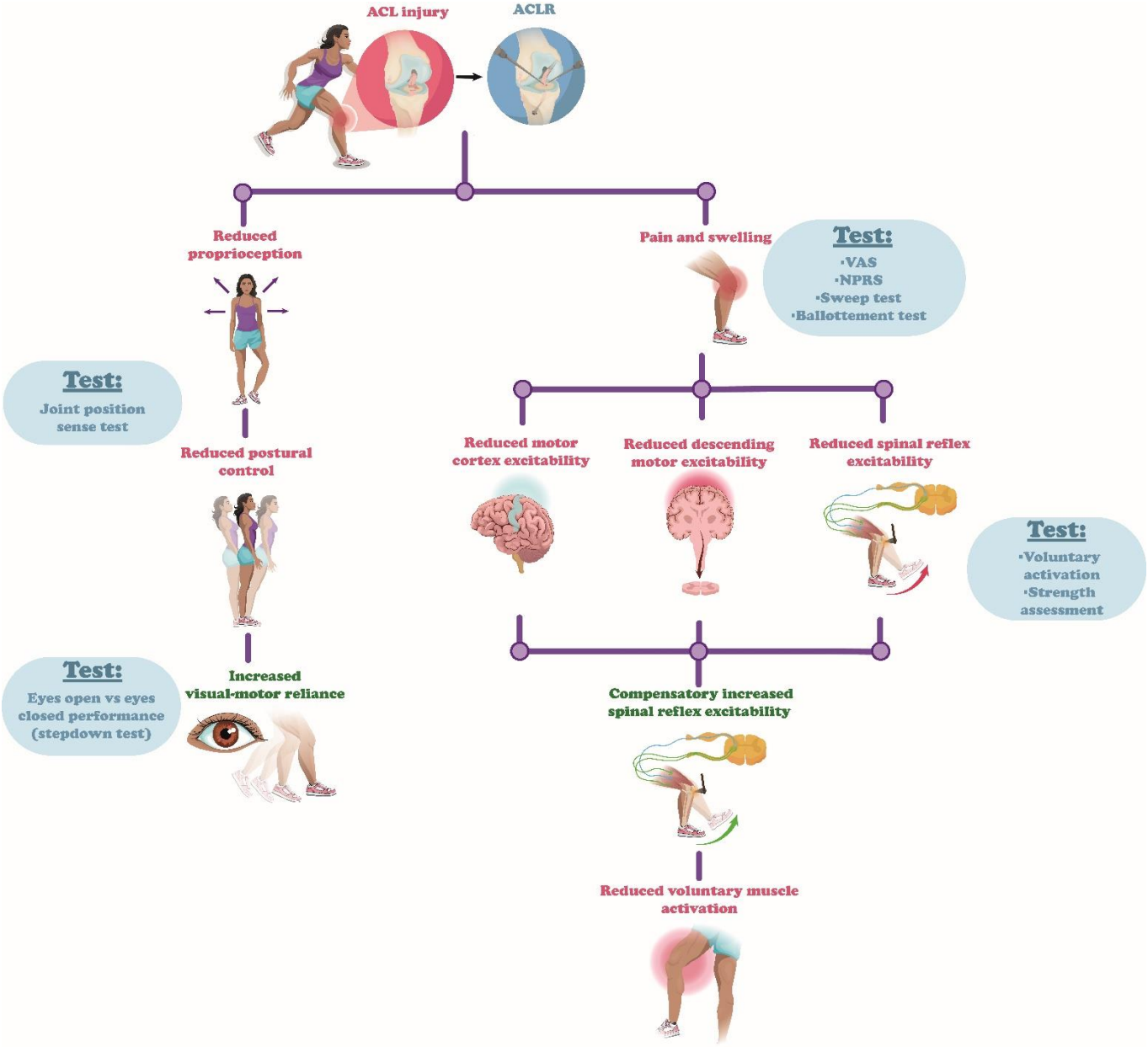

Reduced somatosensory function

Increased visual-motor reli

Reduced peripheral and

FIGURE 1. Sensorimotor dysfunction following ACL injury and ACLR. Also outlines the two priorities for clinicians rehabilitating athletes with sensorimotor dysfunction.

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Improving peripheral and central efferent function

Improving peripheral and central efferent function is important to ensure complete restoration of strength and to avoid developing any compensatory changes to an athlete's movement technique. Peripheral efferent dysfunction is predominantly found in the first few weeks post-injury which reduces spinal reflex excitability. Central efferent dysfunction manifests as descending motor inhibition sub-acutely following surgery.[34,50]

Early-stage rehabilitation after ACLR

Knee joint pain and swelling reduces the spinal-reflexive pathway and quadriceps activation.[30,48] Regular movement of the knee, lower limb strength exercises and gait retraining must be completed early in rehabilitation to help to reduce pain and swelling in the knee.[31] However, additional strategies to improve pain and swelling should be used aggressively in the early stages of rehabilitation to augment recovery. Methods include pain medication, transcutaneous electrical nerve stimulation (TENS) and cryotherapy.[55]

Pain medication strategies can vary. Post-operative pain is generally at its worst in the first three days and nerve blocks are often used to reduce the pain and allow early ambulation and discharge.[7] Nerve blocks reduce post-operative pain and analgesic use.[16] Non-steroidal anti-inflammatory medications are common in outpatient care.[8] They reduce pain and the need for analgesic medication following ACLR.[53] Any medication should be appropriately prescribed by the treating physician and surgeon within the scope of their respective regulatory bodies.

TENS devices apply an electrical impulse to the large diameter afferent nerve fibres via electrodes placed around the injured structure.[51] The afferent information is processed in the central cortex which activates descending inhibitory pathways, resulting in a reduction in peripheral and central hyperalgesia.[19] TENS can reduce knee pain by 36% if the dosage is set to: a strong non-noxious stimulus, low frequency setting of 1-8Hz, high frequency of 25-120Hz, 20 minutes treatment time and at least 5 treatment sessions.[6]

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Cryotherapy reduces pain and the need for analgesic medication.[40,52] Cryotherapy devices, especially those with pneumatic pressure to provide compression to the knee, such as 'Game Ready' (*CoolSystems Inc, USA*) and 'Cryocuff IC Cooler' (*DonJoy Global, USA*), reduce pain significantly more than ice packs alone.[46,62] Cryotherapy-compression significantly reduced participants' pain (by 50%) and led to the cessation of pain medication (83% of participants ceased medication by 6 weeks) as compared to no significant change in pain in those using standard ice packs and only a small portion of these participants ceasing pain medication (23% of participants ceased medication by 6 weeks).[62]

In summary, to improve efferent function in the early stages of rehabilitation, clinicians may choose to recommend regular cryotherapy with compression devices (ideally three times a day or once a day if there are accessibility barriers) for the first 6 weeks post-operative and apply TENS daily for 20 minutes for the first 4-6 weeks.

Throughout rehabilitation

Strength training is a mainstay of ACLR rehabilitation and provides a powerful stimulus to improve peripheral and central efferent function. In the uninjured population, strength training increases motor unit excitation and recruitment, increases corticospinal excitability and reduces intracortical inhibition.[21,28,41] Changes in neural excitability occur within strength training sessions and can be maintained between sessions.[41] Why is efferent dysfunction is measured so commonly in ACLR individuals who have participated in strength training?[34,36,50] Training intensity is a key factor in improving corticospinal excitability.[10] There are significant increases in corticospinal excitability when participants train at 75% of maximum voluntary contraction, and no changes at 25% of maximum voluntary contraction. A progressive resistance program (resistance built up overtime to >90% of 1 repetition maximum) increased the quadriceps and hamstrings strength of ACLR athletes to the same level as healthy controls by 7 months post-surgery.[63] Therefore, training intensity is important during ACLR rehabilitation and previous research may not have trained participants at

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128 sufficiently high intensity to achieve symmetrical strength and to improve their efferent function
129 back to baseline.

130 Neuromuscular electrical stimulation (NMES) and surface electromyography (EMG) biofeedback can
131 be used as adjuncts to strength training to improve efferent function.[11,23] NMES commonly uses a
132 handheld device to apply an electrical current to muscles via electrodes placed on the skin (e.g.
133 'Compex', *DonJoy Global, USA* and 'Neurotrac Sport', *Verity Medical Ltd, UK*). The electrical current
134 elicits a contraction in the muscle. Because athletes in early rehabilitation have diminished spinal
135 reflexive activity and reduced voluntary quadriceps activation,[34] NMES provides an avenue to
136 contract the quadriceps without the risk of heavy loads going through the joint if the athlete was to
137 otherwise perform a high intensity strength exercise. NMES also preferentially recruits type II muscle
138 fibres which may minimise the risk of selective atrophy of the these muscle fibres[37,59] and improve
139 quadriceps force production.[26] NMES in addition to exercises and physiotherapy results in
140 significantly better quadriceps strength when compared to exercises and physiotherapy alone.[13,26,64]
141 NMES was mainly effective when used at high intensities with the contraction time set to 15-20
142 seconds and rest time set to 15-50 seconds.[17,54] NMES was generally used 5-7 days per week for the
143 first 4-12 weeks post-ACLR but the frequency of this may need to be reduced if accessibility barriers
144 exist for athletes.[17,22,54]

145 Surface EMG biofeedback devices are commonly handheld devices, with electrodes placed over the
146 motor units of the target muscles (e.g. 'Neurotrac Simplex', *Verity Medical Ltd, UK* and 'mTrigger',
147 LLC, *Delaware, USA*). They measure the magnitude of motor unit recruitment and provide that
148 information to the user on an easy-to-read display. The clinician can set a target for muscle activity
149 for the athlete to achieve, and the athlete receives an auditory or visual stimulus each time that they
150 achieve the goal. This differs from NMES which is best used as a passive modality. EMG biofeedback
151 training provides an external focus of attention to their training and can improve corticospinal
152 excitability, which is otherwise unaffected by NMES.[49] EMG biofeedback training increases

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quadriceps strength,[12,35] and can be completed with isometric and isotonic exercises.[2,65] A dosage of 5-7 days per week[2,29] is effective in improving strength. However, accessibility barriers for athletes may mean this frequency is not possible. Athletes may find it easier to focus on using EMG biofeedback training during their strength sessions.

In summary (Figure 2), improving efferent function via improving spinal reflexive activity, corticospinal excitability and intracortical excitability is essential throughout rehabilitation. Progressive strength training at a sufficient intensity (>75% of 1 repetition max) to elicit strength and hypertrophy changes, is a powerful stimulus to all components of the efferent system. NMES could be used during the first 6 weeks post-surgery to minimise type II muscle fibre atrophy in the quadriceps[37,59] and to maintain or increase spinal reflexive activity to the quadriceps that would otherwise be inhibited due to the pain and swelling. As the swelling reduces and the athlete can voluntarily activate their quadriceps, reduce NMES to non-strength training days (once or twice a day depending on accessibility barriers of the athlete) and as part of the warmup on a strength session (5-10 minutes in preparation part of session). The athlete may also add EMG biofeedback training to their program during strength sessions until there are no clinically relevant strength or efferent deficit between limbs.

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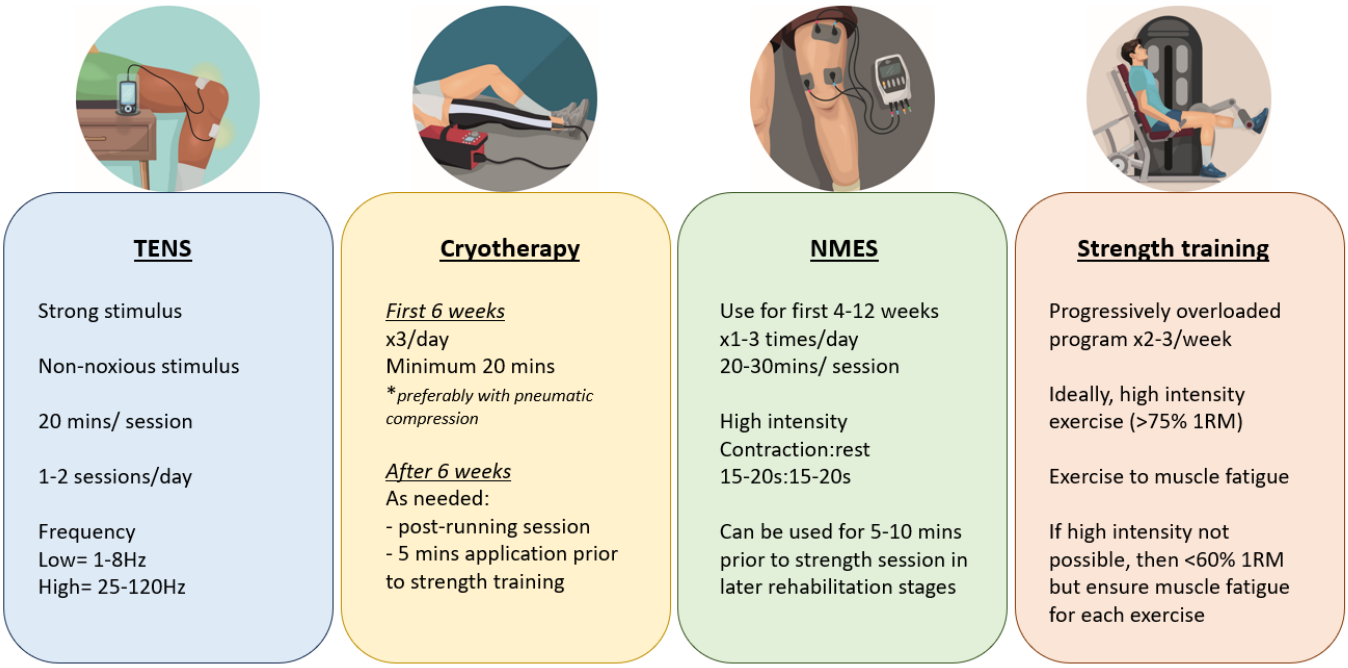

FIGURE 2. Take home messages for clinicians regarding treatment modalities.

Abbreviations: *TENS= transcutaneous electrical stimulation, NMES= neuromuscular electrical stimulation, mins= minutes, Hz= Hertz, s= seconds, RM= repetition maximum*

Improving somatosensory function and reducing reliance on the visual-motor system

Improving somatosensory function early in rehabilitation is essential to restoring normal function, reducing re-injury risk, and minimising reliance on the visual-motor system.[24] Somatosensory dysfunction primarily results from the loss of afferent feedback from the ACL and secondarily from pain and swelling. Therefore, early rehabilitation strategies should focus on improving an athlete's proprioceptive ability and reducing pain and swelling. Regrowth of the proprioceptor nerve fibres into the ACL graft can take months, with no exact time period identified.[4,18] Therefore, improving an athlete's proprioceptive ability in early-stage rehabilitation primarily focuses on improving the control of the surrounding muscles and joints to compensate lack of proprioceptive information from the ACL graft. In the case of non-operatively managed ACL injuries, this focus on control of the

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surrounding muscles and joints needs to be completed throughout rehabilitation as the remaining proprioceptors in the ACL stump may provide altered information.[18,20]

Proprioceptive training focuses on using somatosensory afferent information in the absence of vision.[1] Proprioceptive training significantly improved proprioceptive ability across a range of injuries, including knee pathology.[1] While proprioceptive training reduces the incidence of ACL primary injury in field-based athletes,[14,32,38] there is a lack of research following ACLR to determine its effectiveness.

Implementing proprioceptive training very early in rehabilitation is likely important as reliance on the visual-motor system occurs within the first six weeks after surgery.[33] Proprioceptive training in the clinic predominantly takes the form of balance exercises. Program proprioceptive training such that the complexity of the drill progressively increases. Complexity can be progressed by (Figure 3): (1) the type of task (e.g. stationary standing or a landing task), (2) the amount of visual information (e.g. full vision, blinded vision or partial blinding), (3) the amount of cognitive loading (e.g. dual task loading and cognitive task) and (4) the amount of perturbation (e.g. foam surface or external perturbation application).

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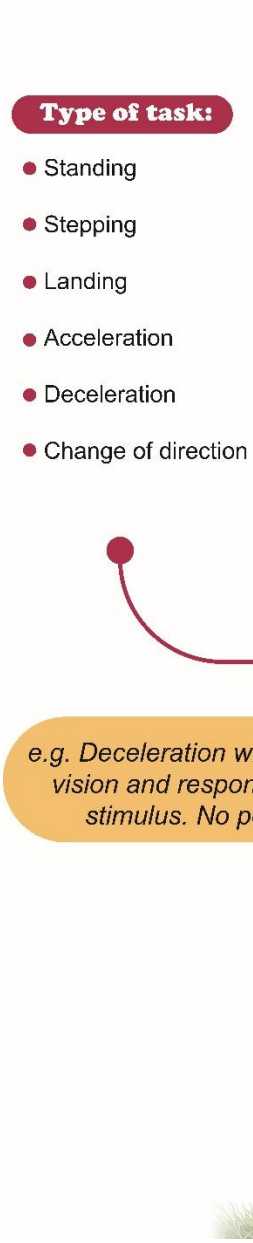

FIGURE 3. Framework outlining factors that need to be considered when creating proprioceptive drills: type of task, visual information, cognitive loading and perturbation.

Factor 1- Type of task

Many of the tasks used for proprioceptive training are low-load and low-velocity in nature, such as single-leg standing or squatting. As rehabilitation progresses, exercises need to progress into higher loads and velocities in multiple movement planes, such as stepping tasks, landing tasks, deceleration tasks and change of direction tasks (Figure 3).

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Factor 2- Visual information

Maintaining postural control or executing a movement requires the athlete to process somatosensory and visual information so that appropriate muscle actions can be executed.[27] If visual information is removed, then the athlete will rely more on proprioception. Clinicians commonly remove vision during a proprioception drill (e.g. standing on one leg eye closed). As the task progresses to a more dynamic task, such as a landing task, being blinded may make the task too difficult to complete or dangerous. A method that can be used to reduce visual information is the use of stroboscopic glasses (e.g. 'Senaptec', *Oregon USA*). Training with stroboscopic glasses facilitates motor skill acquisition.[5]

Stroboscopic glasses flicker between being transparent and opaque so that visual information is partially reduced. The frequency with which the glasses flicker can be altered by clinicians so a single task can have several levels of difficulty. An example is an athlete being tasked to jump and land on a single leg. Initially the task can be completed with a high frequency on the stroboscopic glasses (more visual information available) and as the athlete progresses, the frequency is reduced so that there are longer periods of opacity (less visual information available).

The ideal dosage for stroboscopic training in the ACLR population is unclear. Training dosage in studies investigating skill acquisition vary from short (ten minutes) daily sessions to longer (30-50 minutes) sessions two to three times per week.[3,15,44] We have found three sessions per week with stroboscopic glasses is most practical in exposing athletes to this training. Glasses with pinholes or shutters can substitute for stroboscopic glasses, although may not be as effective.

Factor 3- Cognitive loading

ACL injuries occur in chaotic sporting environments. Athletes need to coordinate movements while processing multiple stimuli (e.g. position of opponents and position on the field) and making multiple decisions (e.g. state of the game, whether to pass and remembering gameplay instructions).

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These factors add to the cognitive burden in athletes and may lead to errors in movement coordination and execution.[56] It is important that clinicians progressively add cognitive load to proprioceptive tasks during rehabilitation.[9,57,58] Cognitive load can be made in the form of: (1) reaction to a stimulus, (2) memory task, (3) decision making task, (4) calculations or (5) a combination of the 4 forms (Figure 3).

Factor 4- Perturbation

Perturbations are unexpected forces applied to the athlete to induce a reactive control of posture.[42] Drills incorporating perturbation should meet two conditions: (1) the external perturbation should induce a rapid and sudden motor response and (2) the perturbation should cause a loss of stability (e.g. a stumble or fall) without a sufficient motor response.[42] Perturbation training is effective in improving balance following ACL injury,[43] chronic ankle instability[45,47] and preventing falls.[39] Examples of drills using varying degrees of perturbation are in Figure 3**.** Examples of how to progress single leg balance, single leg hop and change of direction drills are shown in supplementary videos 1-3.

Summary

The two goals of rehabilitation are: (1) to improve peripheral and central efferent function, and (2) to improve somatosensory function and reduce the reliance on the visual-motor system. There is strong evidence for using strength training, NMES and surface EMG biofeedback to improve efferent function. While these methods are effective in improving efferent function, clinicians must take care to prescribe the interventions at the appropriate intensity to achieve best results. There is good evidence for proprioceptive training, TENS and cryotherapy to improve somatosensory function. However, there is a lack of evidence examining whether these interventions reduce reliance on the visual-motor system. Increasing the complexity of proprioceptive training across type of task,

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amount of visual information, cognitive loading and perturbations may help to improve the effectiveness of proprioceptive training in reducing visual-motor reliance.

Key Points

Findings: To improve sensorimotor function following ACL injury, clinicians could aim to (1) improve peripheral and central efferent function and (2) improve somatosensory function and reduce reliance on the visual-motor system. Interventions can be implemented early in rehabilitation and progressed over time.

Implications: Clinicians can use strength training, NMES and surface EMG biofeedback to improve local and central efferent function. Improving somatosensory function can be achieved through proprioceptive training, TENS and cryotherapy.

Caution: There is a lack of prospective research to identify if proprioceptive training can reduce visual-motor reliance.

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Appendix

Table A1. List of available interventions to target sensorimotor dysfunction following ACLR.

Interventions	Purpose	Equipment needed	Facilitators	Barriers
Pain medication	Allows clinicians to moderate an athlete's pain so that they can progress with rehabilitation	Nil	Wide range of medications that can be effective in reducing pain.	Cost of medication and potential side effects of medications. Could create dependence on medication.
TENS	Electrical pads placed around the knee to deliver a non-noxious electrical impulse to reduce pain from the knee.	TENS machine and pads	Effective in reducing knee pain	Cost of purchasing a device. Time cost with athletes required to use the device regularly
Cryotherapy	Ice packs applied around the knee to cool the skin temperature down. Ideally, this will be used in combination with a pneumatic compression device to be most effective in reducing knee pain.	Ice packs. Ideally a pneumatic compression device as well.	Effective in reducing pain and easy to complete if just using ice packs.	Cost of renting or purchasing the pneumatic compression and cryotherapy device. Time cost with the athletes required to ice regularly in the early stages post-operatively
NMES	Electrical pads placed over the target muscle (usually quadriceps) to deliver an electrical impulse that results in a muscle contraction.	NMES machine and pads	Effective at improving strength and minimising atrophy.	Cost of purchasing a device. Time cost with athletes required to use the device regularly
Surface EMG biofeedback	Electrical pads placed over the target muscle. Athlete is asked to complete an exercise or voluntarily contract the muscle. The device will provide an output	Surface EMG device and pads	Effective in improving central efferent drive and can be paired easily with strength exercises.	Cost of purchasing a device. Time cost with athletes required to use the device regularly

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	as to whether the target muscle activity is achieving the threshold set up the clinician.			
Strength training	A progressively overloaded strength program three times a week and ensuring the athlete works to muscular fatigue will allow clinicians to positively affect an athlete's strength and hypertrophy.	Access to a gym or weights for within the house	Effective in improving strength and hypertrophy following ACLR.	Financial cost of either purchasing weights or gym membership. Accessibility to gyms can be an issue depending on geographical location. Needs appropriate prescription of exercises by taking into account ACLR graft donor site, training age and technique competency.
Proprioceptive training	Using a framework, movement tasks can be manipulated by changing cognitive, visual or perturbation demands to challenge an athlete's proprioceptive ability.	May need items such as blindfolds, strobe glasses, bungee cords, foam pads, etc.	Exercises can be done in any location so athletes can easily complete the exercises.	Cost of purchasing equipment. The most effective training dose post-ACLR is not quite known.

Abbreviations: *TENS*= transcutaneous electrical nerve stimulation, *NMES*= neuromuscular electrical stimulation, *EMG*= electromyography, *ACLR*= anterior cruciate ligament reconstruction